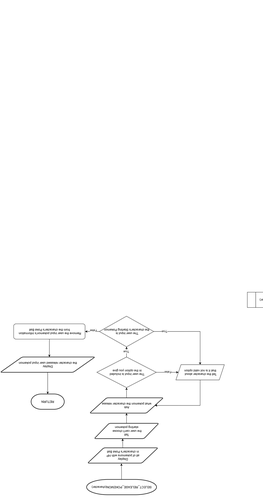

CHECK_BADGE_EILIGIBILITY(character, current_win_count, gym_badge_earned)
level = character's current level
level == 1
win_count = 2
level == 2
win_count = 3
level == 3
win_count = 4
count_left_for_badge = win_count - current_win_count
count_left_for_badge > 0
gym_badge_earned = True
You need to win [count_left_for_badge] more time
RETURN(gym_badge_earned)
ENCOUNTER_STORE(character)
Enter 'buy' to buy a potion, 'use' to use a potion, or 'q' to quit
user_input = buy
BUY_POTION(character)
user_input = use
selected pokemon = SELECT_POTION(character's pokeball)
USE_POTION(character, selected pokemon)
user_input = q
RETURN
Invalid option. Please try again.
SELECT_POKEMON(pokeball)
Display each pokemon and its hp from the pokeball
Select pokemon to proceed by entering the pokemon name
selected pokemon is in pokeball
RETURN(selected pokemon)
Invalid pokemon!
BUY_POTION(character)
Your budget is {the amount of money character has}, and each potion price is {POTION_PRICE}
Enter the number of potions to purchase
number_of_potions > 0
total_price = POTION_PRICE * number_of_potions
Total price will be: {total_price}
budget = the amount of money character has
change = budget - total_price
change >= 0
You can't buy with your current budget
Add the number of potion purchased to the character's potion inventory
Purchase successful! Your remaining budget is {change}
RETURN
You need to buy at least one potion
describe_location(character_status)
Find the cell on the board that matches the character's current location
Display character's current location
GENERATE_STORE_LOCATIONS(board, rows, columns)
count = 0
count < 3
RETURN(board)
row = a random number between 0 (inclusive) and rows (exclusive)
column = a random number between 0 (inclusive) and columns (exclusive)
cordinate of row and column on the board
cordinate of row and column on the board = "Store"
count += 1
IS_ALIVE(character)
alive = True
At least one Pokemon in the character's Poké Ball has HP greater than 0
alive = False
RETURN(alive)
GET_CURRENT_LOCATION(board, character)
current_location = ""
board is empty
RETURN(current_location)
Find the maximum row and column in the board
row_index = 0
row_index < rows
Create an empty row
column_index = 0
column_index < rows
Print row
row_index += 1
location = coordinate of row_index and column_index on the board
location
row += " "
coordinate of row_index and column_index == character's current location
location == "Store"
location == "Gym"
row += "[#]"
row += "[S]"
row += "[G]"
current_location = location
column_index += 1
CURRENT_POKEMON_COLLECTION(character_level, level1, level2, level3)
character_level == 2
current_level_collection = level2
character_level == 3
current_level_collection = level3
current_level_collection = level1
RETURN(current_level_collection)
STARTING_POKEMON_COLLECTION(character_level)
level1_starting_pokemon = level 1 starting pokemon collection that includes pokemon name, type, and currentHP
level2_starting_pokemon = level 2 starting pokemon collection that includes pokemon name, type, and currentHP
level3_starting_pokemon = level 3 starting pokemon collection that includes pokemon name, type, and currentHP
pokemon_collection = CURRENT_POKEMON_COLLECTION (character_level, level1_starting_pokemon, level2_starting_pokemon, level3_starting_pokemon)
RETURN(pokemon_collection)
SET_UP_GAME()
Ask character name
PRINT_INSTRUCTIONS()
character contains data about Money, Current Level, Potion, Current Location, Character Name, Poké Ball, Starting Pokemon
starting_pokemon = STARTING_POKEMON_COLLECTION(character's Current Level)
Display starting pokémon list with their type
Ask which starting pokémon the character wants
user input is in the starting pokémon list
Tell the character about that it is not valid option
character += Put user input pokémon into Poké Ball with its type and HP, and Starting Pokémon
RETURN(character, 1)
PRINT_INSTRUCTIONS()
Welcome to the world of Pokemon! Embark on an exciting journey to become a Pokemon Champion.
Important notes to know before you begin:
Stores are represented by S on the map.
Gyms are represented by G on the map.
If all six of your Pokémon lose their HP, the game is over.
You can only challenge a gym leader once you have a full team of six Pokémon.
After defeating a gym leader, you will earn a badge, unlocking the next level.
Badge Requirements by Level:
Level 1: Defeat the gym leader twice to earn a badge.
Level 2: Defeat the gym leader three times to earn a badge.
Level 3: Defeat the gym leader three times to earn a badge and complete the mission.
The mission is complete when you defeat the final Gym Leader at Level 3.
Your journey will be tough, but with determination and strategy, you can become a Pokémon Champion. Good luck, Trainer!
RETURN
MAKE_BOARD(current_level)
Create an empty board
current_level == 1
rows, columns = 6, 6
Inside the board, mark True for the playable 4x6 rectangle and the 2 extra cells for the start and end, and mark False for all other cells
Mark "Gym" for coordinate of 5 and 5
current_level == 2
rows, columns = 8, 5
Inside the board, mark True for the playable 5x6 rectangle and the 2 extra cells for the start and end, and mark False for all other cells
Mark "Gym" for coordinate of 7 and 4
current_level == 3
rows, columns = 10, 5
Inside the board, mark True for the playable 5x8 rectangle and the 2 extra cells for the start and end, and mark False for all other cells
Mark "Gym" for coordinate of 8 and 4
board = GENERATE_STORE_LOCATIONS(board, rows, columns)
RETURN(board)
CONSTANT
POTION_PRICE = 10
START
character_prev_level = SET_UP_GAME()
board, rows, columns = MAKE_BOARD(character's Current Level)
achieved_goal = False
achieved_goal
Display "Congratulations! You have successfully finished your journey :)"
check = IS_ALIVE()
check
Display "Game Over"
END
prev_level != character's Current Level
board, rows, columns = MAKE_BOARD(character's Current Level)
prev_level += 1
DISPLAY_CURRENT_LOCATION(board, character, rows, columns)
direction = GET_USER_CHOICE(character, board, rows, columns)
is_VALIDATE_MOVE, new_position = VALIDATE_MOVE(board, character, direction)
is_VALIDATE_MOVE
Display character can't go the direction
MOVE_CHARACTER(character, direction, board, rows, columns)
is_special_location = CHECK_IF_CURRENT_LOCATION_IS_SPECIAL(board, character)
is_special_location
EVENT_OCCURRED(character)
achieved_goal = PROCESS_BY_LOCATION_TYPE(character, board)
EVENT_OCCURRED(character)
event_type = SET_EVENT_TYPE()
process_result = True
event_type == No event
event_pokemon_info = DESCRIBE_EVENT(event_type, character)
character_pokemon = TAKE_OUT_POKEMON(character's Poké Ball)
process_result
RETURN
Display what the character's pokemon is with HP
user_choice = SELECT_EVENT_OPTION(event_type, None)
user_choice == Change Pokémon
character_pokemon = CHANGE_POKEMON(character's Poké Ball, character_pokemon)
user_choice == use potion
USE_POTION(character, character_pokemon)
process_result = PROCEED_EVENT_OPTION(user_choice, character_pokemon, character, event_pokemon_info, event_type)
process_result = GET_ATTACKED(event_pokemon_info, event_type, character, character_pokemon)
SET_EVENT_TYPE()
event_collection = wildPokemon, Team Rocket, Strange trainer, No event
event_type = a random event from event_collection
RETURN(event_type)
DESCRIBE_EVENT(event_type, character)
event_pokemon_info = GET_EVENT_POKEMON(character)
Display what type of event occurred and what event pokemon is
RETURN(event_pokemon_info)
GET_EVENT_POKEMON(character)
event_pokemon_collection = EVENT_POKEMON(character's Current Level)
event_pokemon_info = a random pokemon from event_pokemon_collection with its type and HP
event_pokemon_info's pokemon name is in character's Poké Ball
RETURN(event_pokemon_info)
EVENT_POKEMON(character_level)
level1_pokemon = level1 pokemon collection with their type and HP
level2_pokemon = level2 pokemon collection with their type and HP
level3_pokemon = level3 pokemon collection with their type and HP
event_pokemon_collection = CURRENT_POKEMON_COLLECTION(character_level, level1_pokemon, level2_pokemon, level3_pokemon)
RETURN(event_pokemon_collection)
TAKE_OUT_POKEMON(character_pokemons)
player_pokemon = a random pokemon information (pokemon name, type, HP)
player_pokemon's HP == 0
Display what pokemon the user takes out
RETURN(player_pokemon)
MOVE_CHARACTER(character, new_position, board, rows, columns)
character's current location = new_position
DISPLAY_CURRENT_LOCATION(board, character, rows, columns)
RETURN
VALIDATE_MOVE(board, character, direction)
directions = a mapping of directions which contains 1 representing up, mapping to (-1, 0), 2 representing down, mapping to (1, 0), 3 representing left, mapping to (0, -1), and 4 representing right, mapping to (0, 1)
valid_move = False
new_position = None
direction is in directions
RETURN(valid_move, new_position)
dx, dy = direction from directions
new_position = (x-coordinate of character's current location + dx, y-coordinate of character's current location + dy)
Look up for new_position in the board dictionary
new_position exists in the board
valid_move = False
new_position = None
valid_move = the description of new_position in the board
GET_USER_CHOICE(character, board, rows, columns)
Display a numbered list of directions and check the user's status
input_message = "What number would you like to choose (Enter number)? "
error_message = "Invalid input! Please enter a valid number: "
user_choice = CHECK_INPUT_IS_DIGIT (input_message, error_message)
user_choice is not in a numbered list
Display that is not invalid choice
user_choice is check the user's status
Display list of pokemons in character's Poké Ball and how many potions and dollars the user has
DISPLAY_CURRENT_LOCATION(board, character, rows, columns)
RETURN(user_choice)
PROCESS_BY_LOCATION_TYPE(character, board)
achieved_goal = False
current_location = the description of character's current location from board
current_location = Gym
character_has_six_pokemon = HAS_SIX_POKEMONS(character)
character_has_six_pokemon
gym_badge_earned = ENCOUNTER_GYM(character)
gym_badge_earned
character's level is 3
achieved_goal = True
LEVEL_UP(character)
current_location = Store
ENCOUNTER_STORE(character)
RETURN(achieved_goal)
HAS_SIX_POKEMONS(character)
more = 6 - (the number of pokemoons in the character's pokeball)
more > 0
RETURN(not more)
You need to earn {more} more pokemon(s) to enter the gym
ENCOUNTER_GYM(character)
gym_badge_earned = False
Encountered a gym! Enter y to challenge, n to quit
user_input is y or n
Please Enter y or n
user_input is y
Gym Leader: Welcome to the gym! Here is one rule: you can't use potions to accurately assess your skills.
gym_badge_earned = battle_with_gym_leader(character, gym_badge_earned)
RETURN(gym_badge_earned)
BATTLE_WITH_GYM_LEADER(character, gym_badge_earned)
current_win_count, prev_round, gym_round, selected_pokemon, gym_leader_pokemon = INITIALIZE_BATTLE(character)
at_least_one_pokemon_alive = IS_ALIVE(character)
prev_round != gym_round and at_least_one_pokemon_alive and not gym_badge_earned
RETURN(gym_badge_earned)
process_result = True
prev_round += 1
DISPLAY_ROUND_INTRO(gym_round, gym_leader_pokemon, selected_pokemon, character)
process_result is True
user_choice = SELECT_EVENT_OPTION("Gym Leader", gym_round)
process_result, selected_pokemon, stop_process = HANDLE_USER_CHOICE(user_choice, process_result, selected_pokemon, gym_leader_pokemon, character)
current_win_count, gym_badge_earned, gym_round, gym_leader_pokemon = WIN_THE_ROUND(current_win_count, character, gym_badge_earned, gym_round)
stop_process is True
process_result is True
process_result, selected_pokemon, gym_round, stop_process = CHECK_IF_ALIVE_WHEN_LOST_THE_ROUND (gym_leader_pokemon, character, selected_pokemon, gym_round)
stop_process is True
INITIALIZE_BATTLE(character)
current_win_count = 0
prev_round = 0
gym_round = 0
selected_pokemon = TAKE_OUT_POKEMON(character's Poké Ball)
gym_leader_pokemon = GET_EVENT_POKEMON(character)
RETURN(current_win_count, prev_round, gym_round, selected_pokemon, gym_leader_pokemon)
DISPLAY_ROUND_INTRO(gym_round, gym_leader_pokemon, selected_pokemon, character)
❗ Round {gym_round} ❗
Event pokemon status: {gym leader pokemon's name}(HP: {gym leader pokemon's current HP})
{character's name}'s pokemon status: {selected pokemon's name} (HP: {selected pokemon's current HP})
RETURN
HANDLE_USER_CHOICE(user_choice, process_result, selected_pokemon, gym_leader_pokemon, character)
stop_process = False
user_choice = Fight
process_result = FIGHT(selected_pokemon, gym_leader_pokemon, character, "Gym Leader")
user_choice = Change Pokémon
selected_pokemon = CHANGE_POKEMON(selected_pokemon, gym_leader_pokemon, character, "Gym Leader")
user_choice = Run Away
Gym Leader: Running away, huh? I guess today's not your day. Come back when you're ready to battle!
RETURN(process_result, selected_pokemon, stop_process)
CHECK_IF_ALIVE_WHEN_LOST_THE_ROUND (gym_leader_pokemon, character, selected_pokemon, gym_round)
stop_process = False
process_result = GET_ATTACKED(gym_leader_pokemon, "Gym Leader", character, selected_pokemon)
process_result is True
Gym Leader: You lost in round {gym_round}
at_least_one_pokemon_alive = IS_ALIVE(character)
at_least_one_pokemon_alive
stop_process = True
selected_pokemon = TAKE_OUT_POKEMON (character's Poké Ball)
gym_round +=1
RETURN(process_result, selected_pokemon, gym_round, stop_process)
CHECK_IF_CURRENT_LOCATION_IS_SPECIAL(board, character)
is_special_location = False
current_location = the description of character's current location on the board
current_location is Store or Gym
is_special_location = True
RETURN(is_special_location)
GET_STRONGER_COLLECTION(character_level)
character_level == 1
stronger_collection = four number (1, 1.3, 1.4, 1.5)
character_level == 2
stronger_collection = four number (1.3, 1.5, 1.7, 2)
stronger_collection = four number (1.5, 1.7, 2, 2.5)
RETURN(stronger_collection)
MAKE_DAMAGE_STRONGER(event_type, character_level)
stronger_collection = GET_STRONGER_COLLECTION(character_level)
event_type == team rocket
stronger = third number of stronger_collection
event_type == strange trainer
stronger = 1.3
event_type == wild pokemon
stronger = 1
stronger = 1.5
RETURN(stronger)
EVOLVE_POKEMON(character)
evolution_map = a mapping that describes the evolution progression of certain pokemon
current_starting_pokemon_name = character's starting pokemon
current_starting_pokemon_name in evolution_map
evolved_starting_pokemon_name = current_starting_pokemon on the evolution_map
available_starting_pokemons = starting_pokemon_collection that character can get with their current level
evolved_starting_pokemon_name in available_starting_pokemons
RETURN
character's starting pokemon = evolved_starting_pokemon_name
Entirely delete character's Poké Ball info
evolved_starting_pokemon_name in the Poké Ball collection = evolved_starting_pokemon_name in available_starting_pokemons
{current_starting_pokemon_name} has evolved into {evolved_starting_pokemon_name}!
LEVEL_UP(character)
character's current level += 1
EVOLVE_POKEMON (character)
character's current level is 2
character's current location = (0, 0)
character's money += 50
character's current level is 3
character's current location = (0, 4)
character's money += 70
You've leveled up to {character's current level}
RETURN
GET_ATTACKED(event_pokemon_info, event_type, character, character_pokemon)
skill_collection = GET_SKILL_OF(event_pokemon_info's type)
event_pokemon_skill = a random skill from the skill_collection
damage = MAKE_DAMAGE_STRONGER(event_type, character's Current Level)
damage *= a random number from the skill damage range
skill_accuracy = GET_PROBABILITY()
Display what kind of skill the event pokemon used
skill_accuracy
Display the event pokemon's skill missed
Display the event pokemon's skill hit
character_pokemon's HP -= damage
Display how many damage character_pokemon got
process_result = CHECK_STATUS(character_pokemon)
RETURN(process_result)
CHECK_STATUS(character_pokemon)
character_pokemon's HP <= 0
status = True
character pokemon's HP = 0
Display the character pokemon fainted
status = False
RETURN(status)
GET_SKILL_OF(pokemon_type)
skills_of = skill collection depending on type with skill name and damage range
pokemon_skill = skill collection of pokemon_type
RETURN(pokemon_skill)
MAKE_STRONGER(character_level)
character_level == 2
stronger = 1.3
character_level == 3
stronger = 1.5
stronger = 1
RETURN(stronger)
GET_ATTACK_RESULT(character_pokemon_skill, event_pokemon_info, character, event_type)
Display character_pokemon_skill's name hit
damage = MAKE_STRONGER(character's Current Level)
damage *= a random number from the character_pokemon_skill's damage range
event_pokemon_info's HP -= damage
event_pokemon_info's HP <= 0
Display event_pokemon_info's HP, name
event_pokemon_status = True
Display event pokemon is defeated
event_type == Gym Leader
GET_MONEY(character, event_type)
event_pokemon_status = False
RETURN(event_pokemon_status)
GET_PROBABILITY()
accuracy = a random result between True or False
RETURN(accuracy)
GET_MONEY(character, event_type)
earn_money = MAKE_STRONGER(character's Current Level)
event_type == 'wildPokemon'
earn_money *= a random number between 3 and 6
event_type == 'Team Rocket'
earn_money *= a random number between 25 and 30
earn_money *= a random number between 10 and 19
character's Money += earn_money
Display how much money the character got
RETURN
PROCEED_EVENT_OPTION(user_choice, character_pokemon, character, event_pokemon_info, event_type)
user_choice == fight
process_result = FIGHT(character_pokemon, event_pokemon_info, character, event_type)
user_choice == Throw Poké ball
process_result = THROW_POKE_BALL(event_pokemon_info, character)
Display the character escaped from event pokemon
process_result = False
RETURN(process_result)
FIGHT(character_pokemon, event_pokemon_info, character, event_type)
Display event pokemon's name and HP
skill_collection = GET_SKILL_OF(character_pokemon's type)
character_pokemon_skill = CHOOSE_SKILL_TO_CHALLENGE(skill_collection, character's Current Level)
skill_accuracy = GET_PROBABILITY()
skill_accuracy
Display character_pokemon's skill missed
event_pokemon_status = GET_ATTACK_RESULT(character_pokemon_skill, event_pokemon_info, character, event_type)
event_pokemon_status
process_result = True
process_result = False
RETURN(process_result)
CHOOSE_SKILL_TO_CHALLENGE(skill_collection, character_level)
damage = MAKE_STRONGER(character_level)
Display what skills character_pokemon can use with (damage range * damage)
input_message = "Which skill would you like to use (Enter number)? "
error_message = "Invalid input! Please enter a valid number: "
user_choice = CHECK_INPUT_IS_DIGIT (input_message, error_message)
user_choice is not between 1 and 3
Display that is not invalid choice
user_skill = the skill info of the user_choice (skill name and damage range)
RETURN(user_skill)
CHANGE_POKEMON(pokeball, character_pokemon)
Number of the pokeball's pokemon == 1
Display the character has no pokemon to switch to or all other pokemon have 0HP
All other user's pokemon have 0HP
Display the Pokeball's pokemon' status (name, current HP)
Ask Which Pokémon the character wants to switch to
user input is included in the Pokémon you displayed
Tell the character about that it is not valid option
user input pokemon' HP == 0
Tell the character about the pokemon has 0 HP
user input pokemon == character_pokemon's name
Display the character pokemon is switching to
character_pokemon = user input pokemon information (its name, type, currentHP)
RETURN(character_pokemon)
CHECK_INPUT_IS_DIGIT(input_message, error_message)
Display input_message
user input is not digit
Display error_message
RETURN(user_input)
SELECT_EVENT_OPTION(event_type, gym_round)
character_option = CUSTOMIZE_USER_OPTIONS(event_type, gym_round)
Display what option the user can choose
input_message = "What do you want to do (Enter number)? "
error_message = "Invalid input! Please enter a valid number: "
user_choice = CHECK_INPUT_IS_DIGIT (input_message, error_message)
user_choice is not between 1 and number of character_option
Display that is not invalid choice
character_choice = user input's option
RETURN(character_choice)
CUSTOMIZE_USER_OPTIONS(event_type, gym_round)
character_option = Fight, Change Pokémon
event_type == Gym Leader
gym_round == 2
character_option += Run Away
character_option += Use potion
event_type == Wild pokemon
character_option += Throw Poké ball, Run Away
RETURN(character_option)
USE_POTION(character, character_pokemon)
validate_potion = CHECK_POTION(character's Potion, character's Current Level, character_pokemon)
validate_potion
RETURN
Display how many potions the user has
Display character_pokemon's HP
Ask the character wants to use a potion
Tell the character about that it is not valid option
user input == yes
user input == no
pokemon_maximum_hp = LEVEL_MAXIMUM_HP(character's Current Level)
Display how many potions left and what character_pokemon's HP is
character_pokemon's HP > pokemon_maximum_hp - 15
character_pokemon's HP += 15
character_pokemon's HP = pokemon_maximum_hp
character's Potion -= 1
CHECK_POTION(number_of_potion, character_level, character_pokemon)
validation = False
number_of_potion == 0
Display the character has no potion
pokemon_maximum_hp = LEVEL_MAXIMUM_HP(character_level)
character_pokemon's HP == pokemon_maximum_hp
Display character_pokemon has full HP
validation = True
RETURN(validation)
LEVEL_MAXIMUM_HP(character_level)
character_level == 1
maximum_hp = 40
character_level == 2
maximum_hp = 65
maximum_hp = 100
RETURN(maximum_hp)
THROW_POKE_BALL(event_pokemon_info, character)
event_pokemon_info's HP <= 5
Display the character failed to catch the event pokemon
run_or_not = GET_PROBABILITY()
run_or_not
process_result = True
result_checking = CHECK_TOTAL_OF_USER_POKEMONS (character, event_pokemon_info)
result_checking
Display what pokemon the character caught
pokemon_maximum_hp = LEVEL_MAXIMUM_HP(character)
Add the event pokemon to character's Poké Ball with its type and a half of pokemon_maximum_hp
process_result = False
RETURN(process_result)
CHECK_TOTAL_OF_USER_POKEMONS(character, event_pokemon_info)
catch_pokemon = True
number of the pokemons in character's Poké Ball == 6
Display the character can have only up to 6 pokemons
Ask the character wants to release one character pokemon
user input is yes or no
Tell the character about that it is not valid option
user input is no
SELECT_RELEASE_POKEMON(character)
Tell the event pokemon broke free
catch_pokemon = False
RETURN(catch_pokemon)
SELECT_RELEASE_POKEMON(character)
Display all of pokemons with HP in character's Poké Ball
Tell the user can't choose starting pokemon
Ask what pokemon the character release
The user input is included in the option you give
Tell the character about that it is not valid option
The user input is the character's Starting Pokemon
Remove the user input pokemon's information from the character's Poké Ball
Display the character released user input pokemon
RETURN